Software Requirements Specification for Software Engineering: subtitle describing software

Team 24, Maca Boca
Javayria Mudassar
Yilin Jiang
Nour Fawaz
Catherine Yang

October 16, 2026

Contents

1 Introduction **1**
- 1.1 Purpose . . . 2
- 1.2 Scope . . . 2
- 1.3 Stakeholders . . . 2
- 1.4 Definitions and Acronyms . . . 3
- 1.5 Characteristics of Intended Reader . . . 4

2 System Overview **5**
- 2.1 Problem Statement . . . 5
- 2.2 Product Vision . . . 5
  - 2.2.1 Goals . . . 5
- 2.3 System Context . . . 6
- 2.4 User Characteristics . . . 8
  - 2.4.1 Speech-Language Pathologists . . . 8
  - 2.4.2 Researchers . . . 8
- 2.5 System Constraints . . . 9

3 Stakeholder Requirements **9**

4 User Personas **10**
- 4.1 Overworked SLP in Favour of Automating the Speech Analysis Process . . . 10
- 4.2 Detail-Oriented SLP Who Prioritizes Control Over Automated Analysis . . . 11

5 Use Cases **11**
- 5.1 UC1: Log In . . . 12
- 5.2 UC2: Log Out . . . 12
- 5.3 UC3: View Patient List . . . 13
- 5.4 UC4: Select a Submitted Assessment from a Patient . . . 14
- 5.5 UC5: Assign Speaker Roles to the Auto-Generated Transcript . . . 14
- 5.6 UC6: Edit and Update Assessment Results . . . 15
- 5.7 UC7: View Assessment Results and Patient Progress . . . 17
- 5.8 UC8: Review Transcript with Video Playback . . . 17
- 5.9 UC9: Correct a Transcript Chunk . . . 19
- 5.10 UC10: Return Annotated Results to the Patient Record . . . 20
- 5.11 UC11: Generate and Download an Assessment Report . . . 21
- 5.12 Use Case Diagrams . . . 22

6 Functional Requirements **23**
- 6.1 Formal Specification . . . 28
  - 6.1.1 Workflow A: Direct Analysis . . . 28
  - 6.1.2 Workflow B: Annotation-First Analysis . . . 29
  - 6.1.3 Workflow C: On-Demand Refinement . . . 30

7 Data Requirements **31**
7.1 Major Data Entities . . . 31
7.2 Relationships . . . 32
7.3 Annotations . . . 32
7.4 Data Dictionary . . . 33
7.5 Privacy and Data Retention . . . 37

8 Non-Functional Requirements **38**
8.1 Usability . . . 38
8.2 Security . . . 39
8.3 Maintainability . . . 39
8.4 Recording Quality . . . 40
8.5 Accuracy . . . 40
8.6 Performance . . . 40
8.7 Understandability . . . 41
8.8 Compliance . . . 41
8.9 Capacity . . . 42

9 AI and Machine Learning Requirements **42**
9.1 Model Inputs . . . 42
9.2 Model Outputs . . . 42
9.2.1 Example Output . . . 42
9.3 Evaluation Metrics . . . 43
9.4 Known Limitations . . . 43

10 Requirements Traceability Matrices **43**

11 Rationale **44**

12 Likely Changes **45**

13 Unlikely Changes **45**

14 Appendices **46**
14.1 Appendix A: Additional Materials . . . 46
14.2 Appendix B: Development Plan . . . 47
14.3 Appendix C: Values of Auxiliary Constants . . . 47
14.4 Appendix D: Generative AI Disclosure . . . 50
14.4.1 AI Tools Used . . . 50
14.4.2 How and Where Used . . . 50
14.4.3 Verification of AI-Generated Content . . . 51
14.5 Appendix E: Reflection . . . 51
14.6 Nour . . . 52
14.7 Javayria . . . 54

14.8 Catherine 56

Revision History

Date	Version	Notes
14-10-2026	1.0	Added Sections
Date 2	1.1	Notes

[All capstone SRS documents should have a reflection appendix. —TPLT]

[This SRS template is based on the IEEE template, with input from co-pilot and S. Smith. It will get you started. You should not modify the section headings, without first discussing the change with the course instructor. Modification means you are not following the template, which loses some of the advantage of a template, especially standardization. —TPLT]

[Feel free to change the appearance of the report by modifying the LaTeX commands. —TPLT]

[This template document assumes that a single program is being documented. If you are documenting a family of models, you should start with a commonality analysis. —TPLT]

[The SRS is not generally written, or read, sequentially. The SRS is a reference document. It is generally read in an ad hoc order, as the need arises. —TPLT]

[Guiding principles for the SRS document:

- Do not repeat the same information at the same abstraction level. If information is repeated, the repetition should be at a different abstraction level.

—TPLT]

[The template description comments should be disabled before submitting this document for grading. —TPLT]

[You can borrow any wording from the text given in the template. It is part of the template, and not considered an instance of academic integrity. Of course, you need to cite the source of the template. —TPLT]

[When the documentation is done, it should be possible to trace back to the source of every piece of information. Some information will come from external sources, like terminology. —TPLT]

[An SRS document should have the following qualities: unambiguous, consistent, complete, validatable, abstract and traceable. —TPLT]

[The overall goal of the SRS is that someone can learn, understand and verify the captured domain knowledge. They should not have to trust the authors of the SRS on any statements. They should be able to independently verify/derive every statement made. —TPLT]

1 Introduction

[Explain what system is being specified. State purpose, users, and project goals. —TPLT]

[Introduce the name of your software as Software Engineering. Using Software Engineering as a command lets you change the name if necessary and have the change propagate to all documents. —TPLT]

[Include a reference to the template this work is based on. **?** justifies the SRS structure and **?** provides the broader life-cycle context that underlines modern requirements engineering practice. —TPLT]

1.1 Purpose

[State the purpose of the system. —TPLT]

[The purpose should be in the context of the project itself, not in the context of this course. Although the "purpose" of the document is to get a grade, you should not mention this. Instead, "fake it" as if this is a real project. —TPLT]

Software Engineering is intended to solve ... [What problem does your program solve? The description here should be in the problem space, not the solution space. —TPLT]

1.2 Scope

[Briefly describe:

- Problem being solved
- Intended users
- Major functionality
- Expected benefits
- Clarifying the scope often means saying what the software doesn't do

—TPLT]

[The scope section is relevant for understanding the boundaries of the system. It is also relevant for later determining typical values of inputs. The scope should make it clear what inputs are reasonable to expect. This is a distinction between scope and context (context is a later section). Scope affects the inputs while context affects how the software will be used. —TPLT]

TODO

1.3 Stakeholders

The following stakeholders have an interest in the integrated MERLS system.

- **Customer and Project Sponsor:** Dr. Irene Yuan and the research team (Dr. Du, Dr. Wu, and collaborators) who designed the MERLS platform, SAFA, and BeViz. They require an integrated system that supports multiple clinician workflows for analyzing uploaded assessments, is consistent with their existing data collection pipeline, and can be extended in future research.
- **End Users (Speech-Language Pathologists):** The primary users of the system. SLPs review the recorded assessments, evaluate answers, verify and correct auto-generated transcripts, add notes, and use the results to make diagnostic decisions by writing clinical reports.

- **Secondary Users (Researchers):** Speech and language researchers who study assessment data and evaluate how well the technology supports clinicians. They need access to assessments at every stage of analysis in order to determine the accuracy and efficacy of a software.
- **Patients and Guardians:** Bilingual children who complete assessments at home on the patient-facing site, supervised by a parent or guardian. They do not use the clinician side of the system, but they depend on it to keep their recordings private and to be assessed accurately and promptly.
- **Development Team:** Team 24 (Maca Boca), who design, implement, test, and maintain the system during the project. They need clear, testable, and traceable requirements, as well as access to the existing TeleHealth and SAFA code bases.
- **Institutions and Regulatory Bodies:** Clinics, schools, and hospitals that employ SLPs, and regulators of health information (e.g., HIPAA (**?**) for U.S.-based clinicians). They must approve tools that handle patient data and need the system to meet privacy, consent, and data retention obligations.

1.4 Definitions and Acronyms

Term	Definition
MERLS	A web platform for Mandarin-English child language assessment. It brings together the assessment tools clinicians need, so they can complete their entire workflow in one place.
BeViz	Software that organizes and visualizes assessment results. It also detects potential interfering behaviors during an assessment (such as parent prompting) so clinicians can review them.
SAFA	Stands for "Speech Annotation For Application, Access, Accuracy." It automatically generates transcripts from recorded online speech assessments. Clinicians can then review, verify, and annotate the transcripts.
Clinician	A direct user of the system, typically an SLP (speech-language pathologist). Each clinician cares for one or more patients. Clinicians review assessment recordings, verify transcriptions and assessment results, and analyze overall performance.

Patient	An individual who uses MERLS to complete online speech assessments from home. Each patient is assigned to a clinician who oversees their care. Patients can also be referred to as clients.
Report	A detailed breakdown of a patient's results for a single speech assessment completed on the MERLS platform.
Speech Language Pathologist	They can also be referred to as a speech clinician.
HIPAA	Health Insurance Portability and Accountability Act, which is an American federal law that permits the use of sensitive patient health information while protecting their privacy (**?**).
Finite State Machine	A mathematical model that represents a sequential workflow as a set of states and transitions between them. A state may have one or more transitions leading to different states, which allows for multiple pathways through the workflow. At any given moment, the workflow is in exactly one state and may move to another state through a defined transition.
SLP	Speech-Language Pathologist
HCI	Human-Computer Interface
HCD	Human-Centered Design
FR	Functional Requirement
NFR	Non-Functional Requirement

Table 1: Definitions and Acronyms

1.5 Characteristics of Intended Reader

The intended readers of this document are the development team, future developers who maintain the system, the course instructor and teaching assistants, and the supervisor and research team. Readers are expected to have a level of understanding detailed as follows:

- **Software engineering:** Third-year undergraduate knowledge of requirements engineering. This includes reading use cases, functional and non-functional requirements, fit criteria, and finite state machines (the level of McMaster's SFWRENG 3RA3 course).
- **Web development:** Working knowledge of client-server web applications, REST APIs, and cloud storage (e.g., Amazon S3), or enough to relate the requirements to the existing TeleHealth system.

- **Machine learning:** An introductory understanding of automatic speech recognition. This includes what a model's inputs and outputs are and what word error rate measures. Knowledge of training or tuning models is not required.
- **Domain knowledge:** No training in speech-language pathology is required. Clinical terms used in this document are defined in Table 1.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

[The purpose of this section is to provide general information about the system so that the specific requirements in the later sections will be easier to understand. The general system description section is designed to be changeable independent of changes to the functional requirements documented in the specific system description. The general system description provides a context for a family of related models. The general description can stay the same, while specific details are changed between family members. —TPLT]

2.1 Problem Statement

[Current situation, problem, impact, desired improvement. You will likely borrow some text from your problem description document. —TPLT]

TODO

2.2 Product Vision

The purpose of this product is to give speech clinicians a single, centralized telehealth platform for reviewing, annotating, and analyzing speech assessments of bilingual children. By integrating SAFA's annotation tools into the platform and incorporating patient records in one place, the product aims to make the assessment analysis process faster, simpler, and easier to use.

2.2.1 Goals

1. **Integrate SAFA Into the MERLS platform:** Clinicians should be able to annotate assessments, meaning review and correct the written transcript of a recorded session, without leaving MERLS. SAFA is currently a standalone tool, and bringing it into the platform removes the need to switch between systems.
2. **Improve SAFA's Usability:** The annotation interface should be simple and guided enough that any speech clinician can use it with confidence, regardless of technical background.

3. **Speed Up Speech Annotation:** Annotating a bilingual speech assessment should take significantly less time than manual transcription. SAFA has already been shown to reduce annotation time by roughly one-third, and the goal is to preserve or improve on that once SAFA is part of MERLS.

4. **Provide a Clinician Record and Analysis Hub:** Clinicians should be able to find a patient's records and past assessments in a centralized hub. This will simplify clinicians' workflow, allowing them to quickly reference patient history without navigating multiple disconnected systems.

5. **Protect Patient Data:** MERLS handles sensitive health information online; hence the system must keep patient data private and secure. Only authorized clinicians can access a patient's records, recordings, and assessment results.

2.3 System Context

Figure 1 shows the context between the Speech Assessment Web Platform (MERLS) and the users and external systems it interacts with. The platform integrates SAFA into MERLS, which was previously known as TeleHealth. Three types of users will interact with the platform directly, the first being patients and caregivers; they'll complete assessments by submitting video and audio responses, and will receive assessment reports as they progress. Clinicians review transcripts and flagged insights from the system, then edit and annotate them to build reports for their patients. Researchers will correct transcripts to gain insights on how flagging, transcription, and other features can be improved. They also gain data for their research on speech and language patterns. The platform also relies on speech and language models, which generate draft transcripts and detect meaningful insights and patterns within a recording. The database and file storage service is also used to securely store the data that the platform relies on.

[For each of the entities in the system context diagram its responsibilities should be listed. Whenever possible the system should check for data quality, but for some cases the user will need to assume that responsibility. The list of responsibilities should be about the inputs and outputs only, and they should be abstract. Details should not be presented here. However, the information should not be so abstract as to just say "inputs" and "outputs". A summarizing phrase can be used to characterize the inputs. For instance, saying "material properties" provides some information, but it stays away from the detail of listing every required properties. —TPLT]

- Patient and Caregiver Responsibilities:
  - Complete assessments by submitting video and audio responses
  - Record in a setting where the child can be clearly seen and heard
- Clinician Responsibilities:

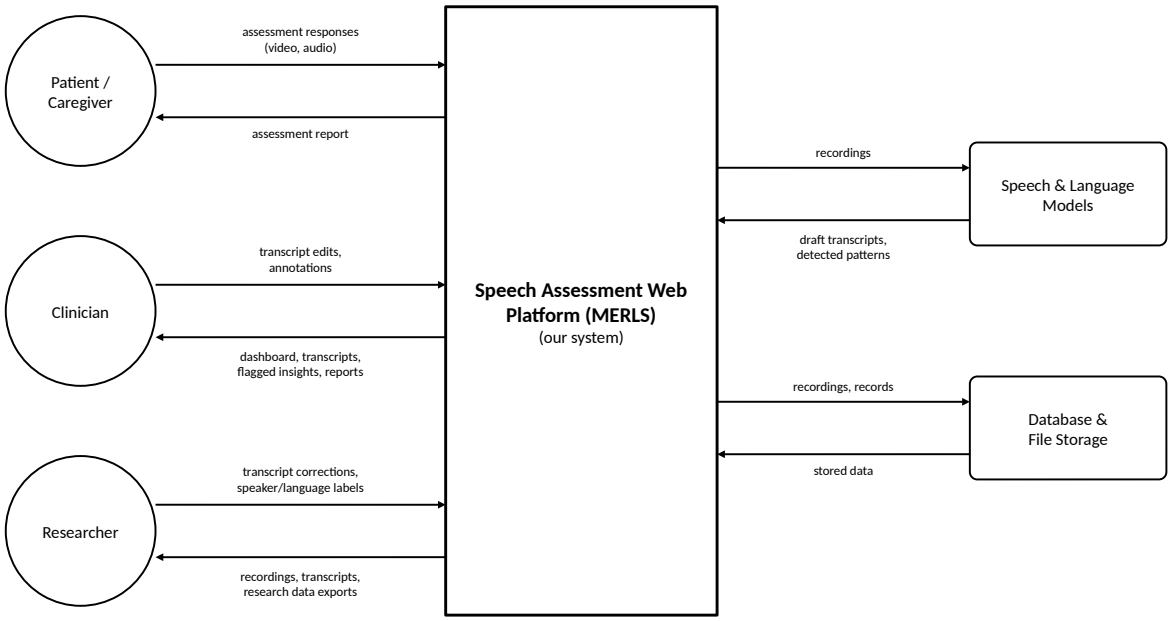


Figure 1: System Context

  - Review flagged items, transcripts, and annotations before relying on them
  - Provide manual evaluations, edits, and notes on assessment results
  - Make the final clinical judgement for each patient

- Researcher Responsibilities:
  - Provide transcript corrections and speaker and language labels
  - Handle exported data according to research ethics requirements

- Speech and Language Model Responsibilities:
  - Return draft transcripts and detected patterns for a given recording

- Database and File Storage Responsibilities:
  - Securely store and return recordings and patient records

- System Responsibilities:
  - Check that submitted recordings meet minimum quality requirements
  - Restrict access to patient data based on the user's role
  - Send recordings to the speech and language models and display the results

– Generate assessment overviews, reports, and data exports

[Identify in what context the software will typically be used. Is it for exploration? education? engineering work? scientific work?. Identify whether it will be used for mission-critical or safety-critical applications. —TPLT]

[This additional context information is needed to determine how much effort should be devoted to the rationale section. If the application is safety-critical, the bar is higher. This is currently less structured, but analogous to, the idea to the Automotive Safety Integrity Levels (ASILs) that McSCert uses in their automotive hazard analyses. —TPLT]

2.4 User Characteristics

The system has two main types of users. The usability requirements (e.g., NFR1) are measured against the typical users described below.

2.4.1 Speech-Language Pathologists

- **Education:** A graduate degree in speech-language pathology and a licence or certification to practice.
- **Experience:** 1 to 10 years of clinical practice with children aged 0 to 18, in schools, hospitals, or private clinics. This is based on the clinicians in the research team's study (Appendix A).
- **Domain knowledge:** Regularly conducts child language assessments (e.g., the Preschool Language Scales) and understands language sample analysis terms such as utterance, morpheme, and code-switching.
- **Language:** Fluent in English. Clinicians who verify another language must also have complete fluency in that language.
- **Technical ability:** Uses everyday office and web software on a desktop or laptop computer. No programming experience or knowledge of speech recognition is expected.

2.4.2 Researchers

- **Education:** A graduate degree, or graduate study, in speech-language pathology, linguistics, or a related field.
- **Technical ability:** Is familiar with spreadsheets, editing tools, and exported data files such as CSV. No programming experience is expected.
- **Role:** Reviews and aggregates assessment data, and does not make clinical decisions.

2.5 System Constraints

- **Integration with MERLS:** The system must be integrated into the existing MERLS telehealth platform and accessed through the clinician dashboard, rather than as a separate standalone application. *Rationale:* Clinician workflows require more than just the analysis of a single assessment. In order to connect their work, they have should a platform that is able to help collect, review, transcribe, and annotate assessments across multiple patients and time.

- **Web-based deployment:** The system must be a web application, deployed to a server and accessible through the web. The SAFA module, which is currently a Python desktop application, must be adapted to run within this web-based environment. *Rationale:* MERLS is a web-based platform, so SAFA must also run in the browser to work within it. This also means clinicians can access the system from any device with a web browser, without needing to install or maintain a separate desktop application.

- **Assessment transcriptions:** For automatic transcription within assessments, the system must use the pre-trained Whisper model under OpenAI, no other models will be trained and used for transcription. *Rationale:* The research team's existing workflow uses Whisper to generate the transcripts reviewed in SAFA, and training a new model is outside the project scope.

- **Health and privacy regulations:** The system must comply with all health and privacy regulations in the country that it's used in (e.g., HIPAA in the United States). *Rationale:* The system stores recordings and records of children, which are protected health information. Leakage of such private information could have serious consequences for these families, as well as legal consequences for the research team.

3 Stakeholder Requirements

This section provides requirements directly stated by our stakeholders in documented interviews, as referenced in Appendix A.

SR1 Clinicians want to spend less time transcribing and annotating recorded assessments, so they can focus on clinical analysis. Clinicians reported that annotating a 20-minute session can take up to four hours, and that the annotation part is "very time consuming".

SR2 Clinicians want to review, annotate, analyze, and report on an assessment in one place, without having to switch between tools or moving files between them.

SR3 Clinicians want a system that "allow(s) the clinician to make the final judgement". Automated transcripts, scores, and flags should act as suggestions that the clinician can review, accept, or correct.

SR4 Clinicians want every result, flag, and transcript segment to be linked to the part of the recording it came from, so they have "proof" to support their clinical decisions.

SR5 Clinicians want to choose how in-depth they would like each assessment to be, whether that be full results first or partial transcription.

SR6 Clinicians want their scores, notes, transcripts, and selected evidence to be compiled into a clinical report for them, instead of assembling the report by hand from a spreadsheet.

SR7 Patients, guardians, and institutions want patient recordings and records to be accessible only to the clinicians responsible for them.

SR8 Clinicians want a system they can learn and use without formal training, regardless of their technical background.

SR9 Clinicians want to analyze assessments that include speech that switches between two languages in order to compare a child's performance in each language.

SR10 Clinicians want to see how a patient's performance changes across assessments over time, as well as the results of a single assessment.

4 User Personas

4.1 Overworked SLP in Favour of Automating the Speech Analysis Process

- Name: James Caveat
- Description: James is a speech clinician at a large hospital, where he is responsible for more than 30 patients. His heavy workload means he must review many speech assessments during his working hours, leaving him little time to carefully absorb and transcribe what he is watching. As a result, it is difficult for him to give each assessment the attention it deserves.
- Goals: Speed up the analysis of speech assessments, receive assessment results quickly, obtain meaningful and clinically useful insights from each assessment, use a single system rather than learning and switching between multiple software tools, and have one simple place to organize his notes and patient information.
- Frustrations: James is discouraged by the amount of time it takes to annotate speech data, exhausted by having to watch and listen to numerous assessment recordings from start to finish, struggles to organize assessment data he collects, and has trouble analyzing large amounts of unstructured data.

- Technical Ability: James is comfortable with the basic tools he uses every day, but he is not technically fluent. He finds it difficult to move between different software systems and prefers to learn one tool well.

4.2 Detail-Oriented SLP Who Prioritizes Control Over Automated Analysis

- Name: Mei Zhang
- Description: Mei is a bilingual speech clinician with eight years of experience who runs a small private clinic. She and her staff deal with select clients that are more complex, where nearly all of them are bilingual children referred for a full diagnostic evaluation. Her main responsibility is deciding whether a child's results indicate a language disorder. Her name is on every report that families, schools, and other professionals rely on. Due to the unique nature of any child's background, she does not accept a result at face value. She wants to see the evidence behind every score and flag, adjust the analysis to each child, and decide for herself how much of the process to automate. She values automation for removing repetitive work, but not for making decisions on her behalf.
- Goals: Choose the level of automation for each assessment, understand and trace automated results to the exact moment in the recording, review and override outputs with a clear record of changes, add clinical interpretations for individual answers and the session as a whole, and build a comprehensive picture of each child's strengths and needs to produce reports she can confidently defend.
- Frustrations: Mei distrusts tools that present results without showing how they were reached, or that do not let her change them. She is frustrated by all-or-nothing automation, where she is forced to either accept an automated output entirely or redo the work from scratch. She worries that automated summaries can hide details that matter clinically, such as an unusual response or an ambiguous sound, and that corrections might overwrite the original output so she can no longer compare them. She is also frustrated when her notes and decisions are scattered across files and tools instead of being kept with the assessment they belong to.
- Technical Ability: Mei is confident with technology and enjoys exploring a tool's features in depth. She is willing to spend time learning a system if it gives her finer control over her work, and she expects to be able to customize how she reviews and organizes information.

5 Use Cases

[Use cases are a way to describe the interactions between users and the system. They are typically written in natural language and describe the steps that a user takes to achieve a

specific goal. Use cases can be used to identify functional requirements and to create test cases. —TPLT] [Typically 8-15 substantial use cases. —TPLT]

5.1 UC1: Log In

Primary Actor

Clinician, Researcher

Preconditions

- User has an existing MERLS account with a clinician or researcher role.
- User is on the platform's login page.

Main Success Scenario

1. User enters their email and password.
2. The system verifies the credentials and authenticates the user.
3. The system identifies the user's role.
4. Using the user's role, the system redirects the user to the correct home page view.

Alternative Flows

1a. The user has forgotten their password: the user selects "Forgot password" and follows the reset process.

2a. When the credentials are incorrect, the system shows an error message and prompts the user to try again.

2b. The user has too many failed attempts, so the system temporarily locks the account until the user resets it.

Postconditions

- The user is logged in, and can only access the data allowed by their role.

5.2 UC2: Log Out

Primary Actor

Clinician, Researcher

Preconditions

- The user is logged in (UC1).

Main Success Scenario

1. The user selects the Log Out option.
2. The system asks the user to save any edits and annotations.
3. The system ends the user's session.
4. The system redirects the user back to the login page.

Alternative Flows

1a. The user is inactive for over IDLE_TIMEOUT minutes, so the system automatically logs them out after saving their work, and shows a message explaining why.

Postconditions

- User cannot access any data or their account again without logging back in.

5.3 UC3: View Patient List

Primary Actor

Clinician

Preconditions

- The clinician is logged in (UC1).

Main Success Scenario

1. The clinician opens the patients tab to view their list of patients.
2. The system retrieves the patients assigned to the clinician.
3. The system displays the list of patients with information, such as their name, age, and the date and status of their most recent assessment.
4. The clinician searches, sorts, or filters the list to find a patient.
5. The clinician selects a patient to view their profile.

Alternative Flows

2a. If the clinician has no assigned patients, the system shows an empty list with a message explaining that no patients are assigned yet.

4a. If no patients match the search or filter criteria, the system shows a "no results" message and the clinician can clear the search.

Postconditions

- Only the clinician can see the patients assigned to them and their information.
- On select, only the selected patient's profile is displayed.

5.4 UC4: Select a Submitted Assessment from a Patient

Primary Actor

Clinician

Preconditions

- The clinician is logged in and has selected a patient (UC3).
- The patient has submitted at least one assessment.

Main Success Scenario

1. The system displays all the patient's submitted assessments, each with its date, assessment type, and review status, which are also filterable.
2. Clinician then selects an assessment.
3. The system displays the assessment's details, including its recordings, metadata, and report if one has been previously submitted.
4. The system provides the option to analyze the assessment, if not previously done.

Alternative Flows

1a. If the patient has not submitted any assessment, the system shows a message saying no assessments are available.

3a. The recording is missing or corrupted, so the system shows an error and flags the assessment to be followed up on.

Postconditions

- The selected assessment is ready to be analyzed and annotated with SAFA.

5.5 UC5: Assign Speaker Roles to the Auto-Generated Transcript

Primary Actor

Clinician, System

Preconditions

- The clinician must be logged into their account in the system
- The clinician has selected a patient and opened one of that patient's video assessments

Main Success Scenario

1. The clinician selects and loads the video assessment they want to analyze
2. The system auto-generates transcriptions for the video assessment

3. The system displays the video playback along with the auto-generated transcriptions as chunks of text
4. The clinician selects a chunk
5. The system indicates which chunk is selected and presents that chunk's details for editing
6. The clinician selects the "Speaker" dropdown in the chunk's detail view
7. The system displays the following list of options for different possible speakers
    - System
    - Child
    - Parent
    - Experimenter
    - Multiple Speakers
    - Others
8. The clinician selects the correct role for the current speaker
9. The system labels the chunk with the selected role
10. The clinician repeats steps 4-9 for the remaining chunks

Alternative Flows

10a. The clinician may select any segment in any order, and the system indicates the newly selected segment. The clinician can then make any edits to the chunk of text using the given options.

Postconditions

- Each segment the clinician labeled has its assigned speaker role, and the roles are saved with the transcript.
- The labeled roles are still shown when the clinician reopens the assessment.

5.6 UC6: Edit and Update Assessment Results

Primary Actor

Clinician

Preconditions

- The clinician has an account and is logged in

- The clinician is analyzing one of their patient's speech assessments

Main Success Scenario

1. The clinician opens the assessment questions for the assessment
2. The system displays the assessment questions alongside the video of the point in the recording where each question was asked
3. The system indicates which questions have potential interference behavior
4. The clinician selects a flagged question
5. The clinician decides the flag is unjustified and removes it
6. The system saves the change and shows the question without the flag
7. The clinician moves on to another part of the analysis, such as annotating the transcript
8. The clinician notices interference behavior from a parent in a part of the recording they had previously cleared
9. The clinician returns to the affected question
10. The clinician flags the question again and adds a note describing the parent interference
11. The system saves the flag and notes and shows them with the question

Alternative Flows

3a. The system does not flag a question, but the clinician notices interference while working on another part of the analysis. The clinician returns to that question, flags it, and adds a note.

10a. The clinician changes their mind again and removes the flag or edits the note. The system saves the update and shows the latest version.

Postconditions

- The question shows the clinician's latest flag status and notes
- The latest flag status and notes are included in the generated assessment report.

5.7 UC7: View Assessment Results and Patient Progress

Primary Actor

Clinician

Preconditions

- The clinician has an account and is logged in
- The clinician has a patient with at least one completed assessment

Main Success Scenario

1. The clinician selects a patient from their list of clients
2. The system displays an overview of the patient, showing their performance trends across attempts and a list of their completed assessments
3. The clinician selects an assessment from the list
4. The system displays a summary of that assessment session, including:
    - The date of the assessment
    - The overall score
    - The recording of the session
    - The list of questions asked

Alternative Flows

3a. The clinician wants to see how the patient's performance has changed over time. Hence, the clinician reviews the trends shown in the patient overview, where the system presents the patient's results across multiple attempts.

Postconditions

- The clinician has seen a single assessment's results as a session summary along with the patient's progress across different attempts as a trend.

5.8 UC8: Review Transcript with Video Playback

Primary Actor

Clinician, Researcher

Preconditions

- The user is logged into their account in the system
- The user has selected and opened a patient's video assessment that has an auto-generated transcript

Main Success Scenario

1. The user opens the transcript of the assessment
2. The system displays the video playback alongside the transcript, divided into time-coded segments, and marks the start of each segment on the playback timeline
3. The user starts playback
4. The system highlights the segment being spoken at the current playback position and scrolls the transcript to keep it visible
5. The user notices a segment they want to review more closely and selects it
6. The system moves playback to the start of the selected segment and highlights it
7. The user replays the segment as many times as needed
8. The user pauses playback once they have finished reviewing
9. The user closes the transcript if they would like

Alternative Flows

2a. The transcript has not finished generating. The system displays the video playback and indicates that the transcript is still being generated. The use case resumes at step 2 once the transcript is ready.

3a. The user selects a segment before starting playback. The system moves playback to the start of that segment, and the use case continues at step 4 when the user starts playback.

4a. No speech occurs at the current playback position. The system does not highlight any segment of the transcript until playback reaches the next segment.

Postconditions

- The highlighted segment matches the current playback position.
- The transcript and assessment results are unchanged.

5.9 UC9: Correct a Transcript Chunk

Primary Actor

Clinician, Researcher

Preconditions

- The user is logged into their account in the system
- The user is reviewing an assessment's transcript alongside its video playback (UC8)

Main Success Scenario

1. The user selects a segment they determine includes inconsistencies (text does not match what is said in the recording)
2. The system moves playback to the start of the segment, highlights it, and displays all additional details. The system indicates that the segment has not yet been verified.
3. The user replays the segment
4. The user corrects the text of the segment
5. The user changes the segment's language tag to reflect the language spoken (can be more than one)
6. The user saves the changes
7. The system saves the edited segment, keeps the original auto-generated version, and displays the differences between the two
8. The system marks the segment as verified and records who made the change and when

Alternative Flows

1a. The user selects a segment that is already correct. The user marks it as verified without editing it, and the system marks it as verified and records who verified it and when.

6a. The user decides to discard their edits. The system restores the segment to its last saved version.

6b. The user tries to leave the transcript with unsaved changes. The system asks the user whether to save or discard them before leaving.

8a. The user later reverts a verified segment to its original auto-generated version. The system restores the original text and metadata, and records the change.

Postconditions

- The segment shows the user's corrected version and is marked as verified.

- The original auto-generated version of the segment is preserved and can be compared with the corrected version.
- The system has recorded who changed the segment and when.

5.10 UC10: Return Annotated Results to the Patient Record

Primary Actor

Clinician

Preconditions

- The clinician is logged into their account in the system
- The clinician has reviewed and corrected the transcript of one of their patient's assessments (UC9)

Main Success Scenario

1. The clinician indicates that they have finished annotating the transcript
2. The system shows how many segments have been verified and how many have not
3. The clinician confirms that the annotation is complete
4. The system saves the annotated transcript to the patient's assessment record and marks the transcript as verified
5. The system returns the clinician to the assessment's results
6. The system shows the annotated transcript as part of the assessment, alongside its results and notes

Alternative Flows

2a. Unverified segments still remain. The clinician chooses to return to the transcript, and the system moves to the first unverified segment. The use case resumes at step 1 once the clinician finishes.

3a. The clinician completes the annotation with unverified segments remaining. The system saves the transcript to the record and indicates which segments are still unverified.

4a. The clinician annotated only a selected portion of the recording (on-demand refinement). The system saves the annotated segments to the assessment record and returns the clinician to the assessment question they were reviewing.

4b. The system cannot save the annotation, for example because the network connection was lost. The system keeps the clinician's work, informs the clinician that the save failed, and lets them try again.

Postconditions

- The annotated transcript is stored in the patient's assessment record without the clinician exporting, transferring, or importing any files.
- The annotated transcript is shown when the assessment is opened from any of the clinician's views, and is used when the clinician generates the report.

5.11 UC11: Generate and Download an Assessment Report

Primary Actor

Clinician

Preconditions

- The clinician is logged into their account in the system
- The clinician has reviewed one of their patient's assessments, using any of the three analysis workflows

Main Success Scenario

1. The clinician chooses to generate a report for the assessment
2. The system compiles the report from the assessment's current results, including:
    - The patient and assessment details, such as the date, language, and assessment type
    - The overall score and the evaluation of each question
    - The interference flags the clinician kept
    - The clinician's notes on each question and on the assessment as a whole
    - The annotated transcript
3. The system displays a preview of the report
4. The clinician reviews the preview
5. The clinician downloads the report
6. The system provides the report as a PDF file and records the download

Alternative Flows

1a. A report has already been generated for the assessment, and the assessment has not changed since. The system displays the existing report, and the use case continues at step 4.

1b. The assessment has changed since its report was generated. The system indicates that the report is out of date, and the clinician chooses to generate a new one. The use case continues at step 2.

2a. The transcript has not been annotated, or some segments have not been verified. The system includes the automatically generated text and marks it in the report as unverified.

4a. The clinician notices an error in the results while reviewing the preview. The clinician returns to the assessment, corrects the result, and generates the report again.

5a. The clinician only wants to view the report. The clinician closes the preview, and the report remains available from the assessment.

Postconditions

- The report reflects the assessment's results, flags, notes, and transcript at the time it was generated, including every change made by the clinician.
- The report is available from the assessment and can be viewed or downloaded again later.
- The download is recorded with the clinician and the time.

5.12 Use Case Diagrams

The following use case diagram shows the interactions between the clinician and the system for certain use cases described above.

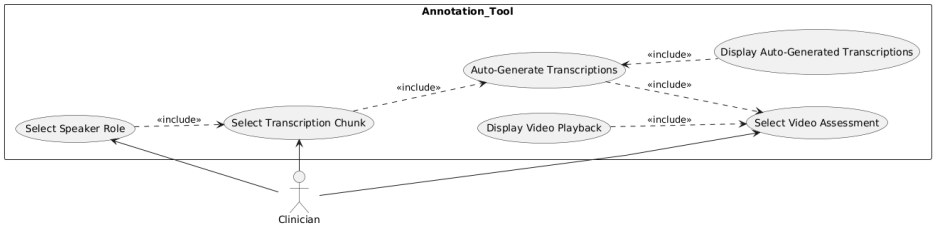


Figure 2: Use case diagram showing how a clinician assigns speaker roles to an assessment transcript using the annotation tool in MERLS. A detailed description is given in Use Case 5.

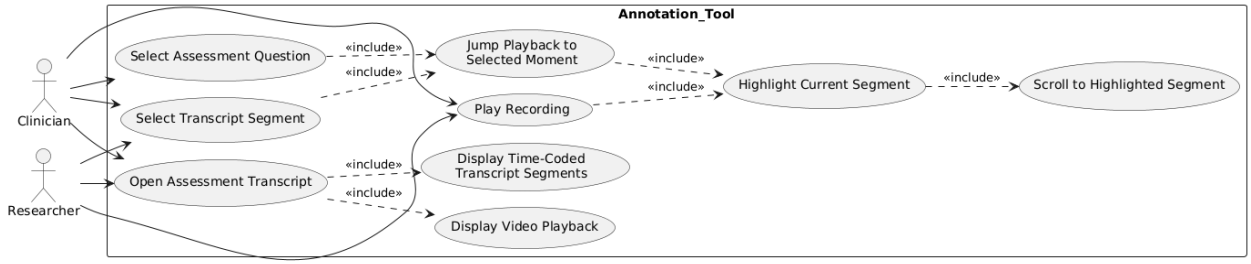


Figure 3: Use case diagram showing how a clinician or researcher reviews an assessment transcript alongside its video playback in MERLS, including the synchronized highlighting of transcript segments and jumping to a selected segment or question. A detailed description is given in Use Case 8.

6 Functional Requirements

FR1 : The system shall ...
Rationale:
Fit Criterion:

FR2 : The system shall ...
Rationale:
Fit Criterion:

FR3 : The system shall ...
Rationale:
Fit Criterion:

FR4 : The system shall ...
Rationale:
Fit Criterion:

FR5 : The system shall ...
Rationale:
Fit Criterion:

FR6 : The system shall allow clinicians to assign speaker roles and filter assessment transcripts by speaker.
Rationale: Automatically generated transcripts do not identify who is speaking. Clinicians need to distinguish the child's speech from that of adults, such as the parent or clinician, to analyze the child's language and identify interfering parent behavior.
Fit Criterion: The system must label every transcript segment from a speaker with the role assigned to that speaker once the clinician assigns it.

FR7 : The system shall generate detailed assessment reports from transcriptions, including any edits made by clinicians.
Rationale: Clinicians review and correct the automatically generated transcription, adjust flagged assessment questions, and add notes to individual questions. The report must reflect these changes so that it includes the clinician's judgment.
Fit Criterion: The system must include any changes made throughout the analysis process made by the clinician, otherwise it outputs the automatically generated content in the report.

FR8 : The system shall allow clinicians to edit and update assessment results at any time.
Rationale: Clinicians do not always analyze assessments sequentially. They may notice an error in a result or change their judgment after reviewing more of the recording, so they need to be able to go back and change results at any stage.
Fit Criterion: The system must display the updated result once the clinician submits a change.

FR9 : The system shall display a list of all assigned patients to the logged-in clinician.
Rationale: Clinicians care for multiple patients and need access to their information to properly care for each one. Additionally, they need a quick starting point for their work.
Fit Criterion: The system must display every patient assigned to the clinician once they log in. The system must display an empty list when the clinician has no assigned patients.

FR10 : The system shall allow clinicians to add new patients.
Rationale: Clinicians regularly take on new patients and need to create a profile for each one so that the patient's assessments and records can be managed in the system.
Fit Criterion: The system must add the new patient to the clinician's list of patients once the clinician creates the profile.

FR11 : The system shall display to the clinician all completed assessments for a selected patient, ordered from most recent to oldest.
Rationale: Clinicians must be able to track a patient's progress across multiple assessments over time.
Fit Criterion: For a selected patient, all completed assessments are listed, ordered from most recent to oldest by assessment date.

FR12 : The system shall display an overview of every assessment's results, including automatically flagged items and the patient's overall results, without requiring a transcript.
Rationale: When reviewing many assessments, clinicians need a quick summary without having to perform any in-depth transcription and annotation.
Fit Criterion: The overview for each completed assessment shows the patient's overall results, and each question's result (e.g., right or wrong) and whether it was flagged. The overview can be viewed before the assessment is reviewed in the SAFA tool.

FR13 : The system shall flag questions in video assessments where potential interference occurred.
Rationale: During assessment, outside help or disruptions (e.g., parents giving hints to the child) can heavily impact results and potentially make them invalid.
Fit Criterion: Each question with detected interference is visibly marked in the assessment overview.

FR14 : The system shall allow the clinician to navigate to the point in the video assessment where a selected question was asked or a selected moment occurred.
Rationale: When searching for a specific moment or question asked in an assessment, when a transcript is available, clinicians should not have to search through a full recording to review that moment.
Fit Criterion: Selecting a question or specific moment in a conversation starts video playback at that specific point in the assessment.

FR15 : The system shall allow the clinician to manually evaluate questions that cannot be automatically evaluated, using video playback of the recorded answer.
Rationale: Some responses, such as open-ended answers, cannot be evaluated automatically and require clinical judgement. Other times, the system simply might not be able to automatically evaluate a question, so the clinician must manually evaluate it.
Fit Criterion: For each question without an automatic result, the clinician can play the recorded answer and enter their evaluation, which is then saved to the assessment results.

FR16 : The system shall allow clinicians to manually evaluate assessment questions that cannot be graded automatically, while playing back the portion of the recording in which the question was answered.
Rationale: Some question types, such as repetition questions where the child repeats a sentence, have no single correct option that the system can compare against. The clinician must listen to and watch the child's response to judge it, and the recording is the evidence for that judgment.
Fit Criterion: For every question that is not automatically graded, the system must let the clinician play the recording segment for that question and record an evaluation (e.g., correct or incorrect). The saved evaluation must be shown with the question and included in the assessment's score.

FR17 : The system shall allow clinicians to add, edit, and delete a note on each individual assessment question.
Rationale: Clinicians record observations about specific answers, such as the type of error made or potential speech clues for that question's response. These observations support the final evaluation and must stay attached to the question they describe.
Fit Criterion: After the clinician saves a note on a question, the system must show

that note with the question every time the assessment is reopened, unless the clinician edits or deletes it.

FR18 : The system shall allow clinicians to add, edit, and delete notes for an entire video assessment.
Rationale: Some observations apply to the whole session rather than a single question, such as the child's attention, the home environment, or the recording quality. Clinicians need a place to record these alongside the assessment.
Fit Criterion: After the clinician saves a note on an assessment, the system must show that note on the assessment's view every time the assessment is reopened, unless the clinician edits or deletes it.

FR19 : The system shall store completed transcript annotations in the patient's assessment record.
Rationale: SAFA currently runs as a standalone tool, and its annotations exist only in files that clinicians export and move by hand. Annotations need to stay linked to the assessment and patient they belong to, so they can be reviewed later and used in reports.
Fit Criterion: After a clinician saves an annotation, the annotation must appear in that patient's assessment record when it is opened from any of the clinician's views, with no file upload or download step performed by the clinician.

FR20 : The system shall highlight the transcript segment that corresponds to the current playback position of the assessment recording.
Rationale: Clinicians verify transcripts by watching and listening to the recording. Highlighting the segment being spoken lets them follow along without searching for their place in the transcript.
Fit Criterion: While the recording plays, the highlighted segment must be the segment whose start and end times contain the current playback position, and the transcript must scroll so that the highlighted segment is visible.

FR21 : The system shall move the recording's playback position to the start of a transcript segment when the clinician selects that segment.
Rationale: Clinicians often need to re-watch a specific moment, such as a child's answer or a flagged interaction, to verify the transcript or their evaluation. Jumping directly to it avoids scrubbing through the recording by hand.
Fit Criterion: After the clinician selects a segment, the playback position must be set to the segment's start time, and that segment must be highlighted.

FR22 : The system shall display assessment results in a format suited to the assessment's purpose.
Rationale: Clinicians use progress-monitoring assessments to track change over time and diagnostic or screening assessments to evaluate a patient at a single point in time. Each purpose requires a different view of the results to be interpreted correctly.

Fit Criterion: The system must show progress-monitoring results as change over time and diagnostic or screening results as a single-session summary.

FR23 : The system shall organize assessment results by clinically meaningful domains, such as semantics, grammar, and phonology.
Rationale: Clinicians interpret a child's language abilities by domain, so grouping results this way supports diagnosis and reporting.
Fit Criterion: Every question in an assessment is assigned to a domain, and the clinician can view results grouped and summarized by domain.

FR24 : The system shall ...
Rationale:
Fit Criterion:

FR25 : The system shall provide breaks during assessments to keep children engaged.
Rationale: Children with limited attention or complex communication needs may disengage during long screen-based assessments.
Fit Criterion: The system presents an interactive break activity (e.g., an animation) after a set number of questions defined for each assessment, after which the assessment resumes where it left off.

FR26 : The system shall visualize a patient's assessment results by clinical domain (areas of speech) and language.
Rationale: Clinicians who tested the previous platform valued charts and percentages, and asked for results by language to be shown side by side and grouped into domains such as semantics, grammar, and phonology, to help tell a language difference from a disorder. This requirement is outside the current project scope and is linked to the Support Multiple Assessment Comparison stretch goal from the Problem Statement.
Fit Criterion: For an assessment completed in more than one language, the system must display the results for each clinical domain in a single, consolidated view.

FR27 : The system shall automatically identify potential grammatical and phonological errors in the assessment transcript and present them to the clinician as suggestions for review.
Rationale: Clinicians see transcription as an intermediate step, and they asked for the system to identify grammatical and error patterns so they can focus on interpretation. Since ambiguous speech (e.g., a dialect feature versus an articulation error) can be clinically meaningful, detections must remain suggestions that the clinician confirms. This requirement is outside the current project scope.
Fit Criterion: Each detected error must be linked to its transcript segment and shown as unverified until the clinician accepts or rejects it. Only accepted errors may appear in the assessment report.

6.1 Formal Specification

6.1.1 Workflow A: Direct Analysis

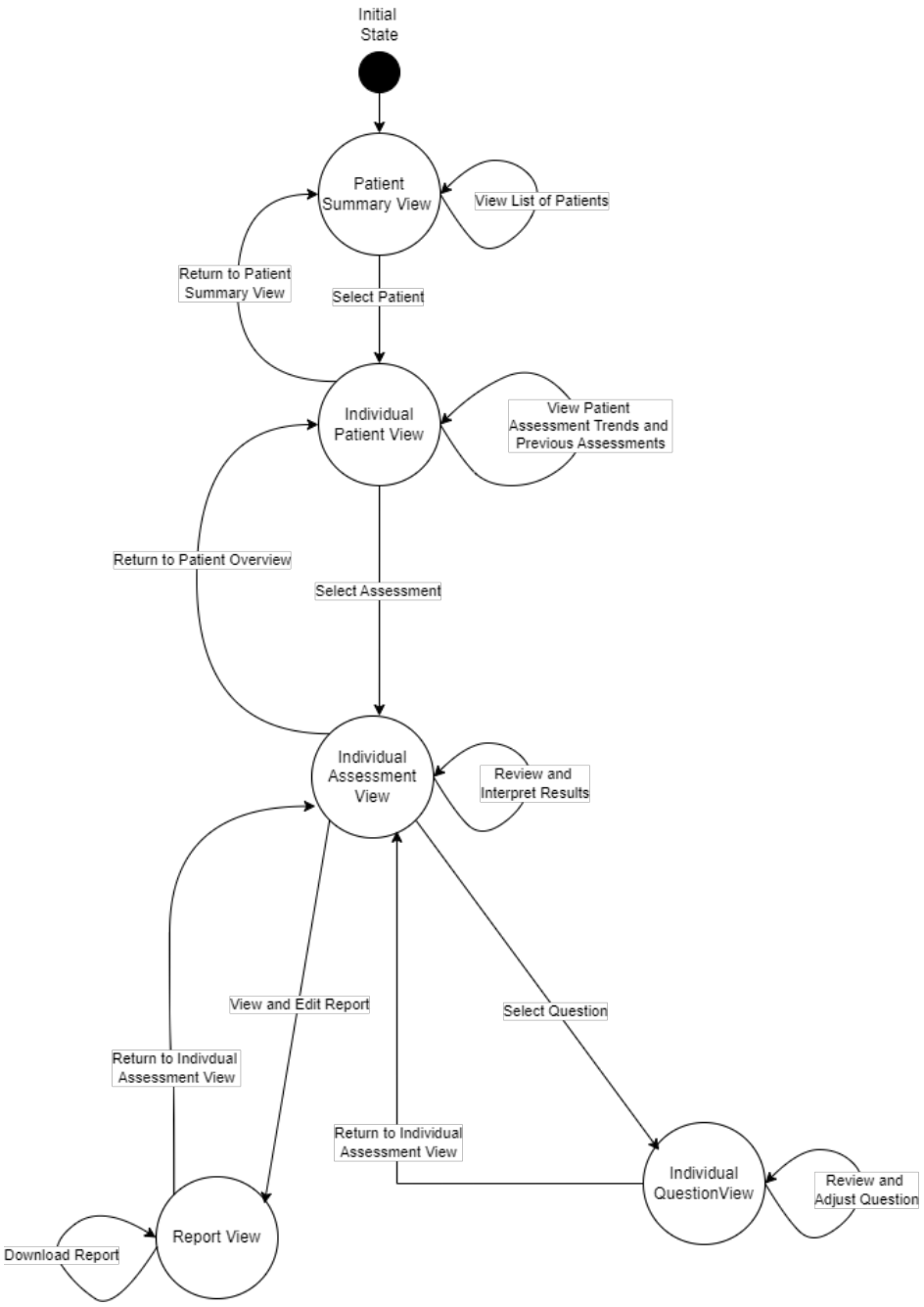


Figure 4: A finite state machine illustrating the direct analysis workflow (Workflow A) of a clinician when using the MERLS system.

6.1.2 Workflow B: Annotation-First Analysis

In this workflow, the clinician chooses to annotate the assessment recording before reviewing the results. The clinician is provided the ability to edit and correct the auto-generated transcript in the annotation view, where the annotated transcript is then saved to the assessment record. The clinician then reviews the questions and results, and prepares the report, using the annotated transcript as evidence.

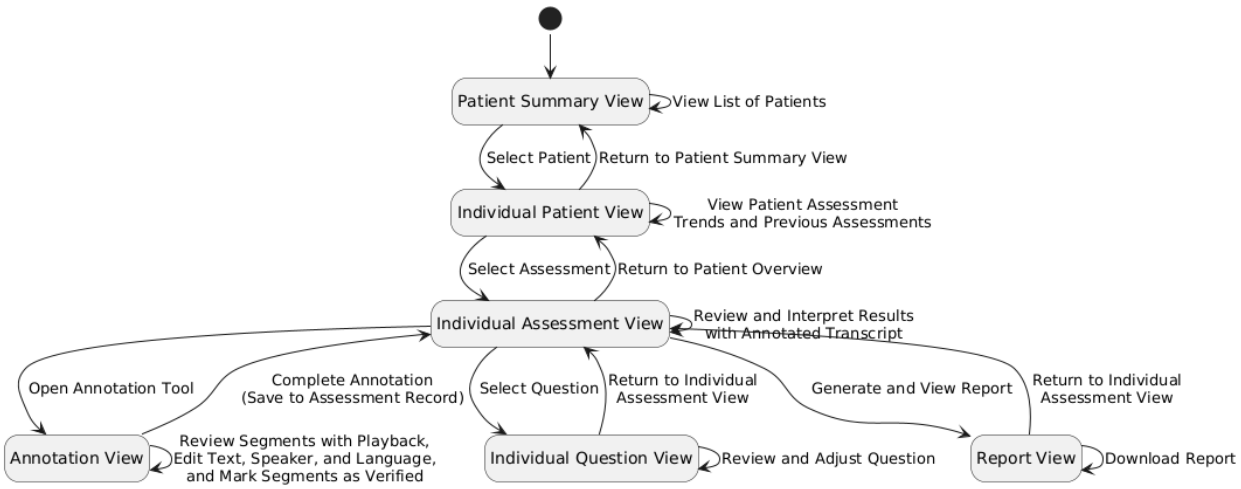


Figure 5: A finite state machine illustrating the annotation-first analysis workflow (Workflow B) of a clinician when using the MERLS system.

6.1.3 Workflow C: On-Demand Refinement

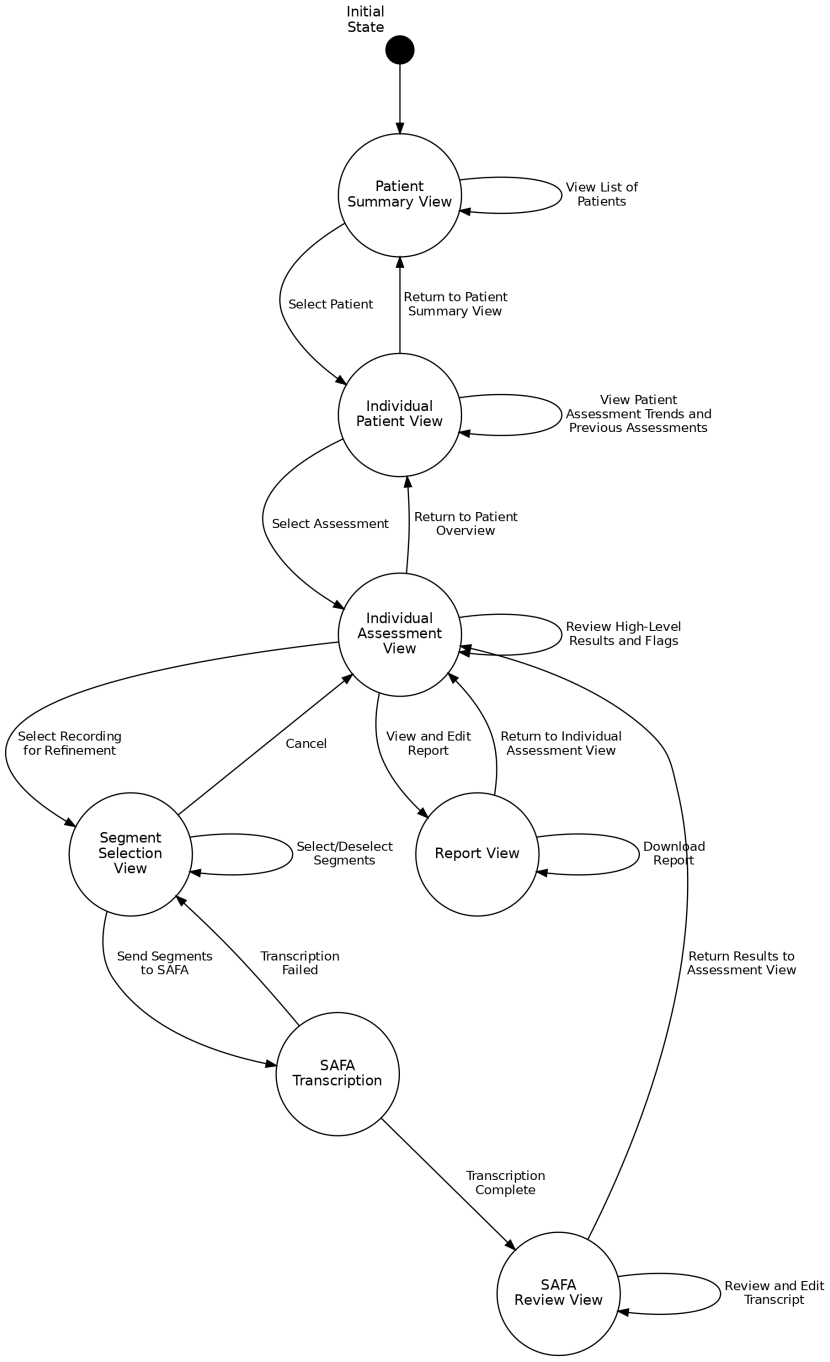


Figure 6: A finite state machine illustrating the on-demand refinement workflow (Workflow C) of a clinician when using the MERLS system.

7 Data Requirements

This section comprises the major data entities and data dictionary below.

7.1 Major Data Entities

Table 2 lists the information the system must manage, grouped by its role. Their relationships are described in the next section.

Entity	Description
People	
User Account	The account a clinician, researcher, or administrator uses to sign in to the system. It determines what information the user may access.
Clinician	A speech-language pathologist who uses the system to review their patients' assessments.
Patient	A child being assessed. Since patients are minors, the patient record also identifies their guardian (a parent or caregiver), who consents to the recording of assessments and acts on the patient's behalf in privacy matters.
Assessment content	
Question	A test item in the question bank, such as a matching or repetition item, which is the same for every patient.
Assessment	A single test session completed by a patient in one language. Its results are made up of its question responses, the overall score, and the clinician's notes.
Question Response	A patient's answer to one question in an assessment, including the recording of a spoken answer, together with the clinician's evaluation and note for that question.
Recording and annotations	
Recording	The video and audio recording of the whole assessment. Question responses, transcript segments, and interference flags all refer to points or periods on this recording's timeline.
Transcript	The written record of the speech in a recording.
Transcript Segment	A time-coded part of a transcript, holding both the automatically generated version and the clinician's edited version.
Interference Flag	A moment in a recording where possible interference, such as parent prompting, was detected by the system or noted by the clinician.
Assessment Note	A clinician's note on an assessment as a whole.

Entity	Description
Records	
Report	A clinical report summarizing a single assessment.
Audit Log Entry	A record of a single view, change, download, or export of patient information (NFR8).

Table 2: Major Data Entities

7.2 Relationships

The relationships between the data entities are as follows:

- Each clinician, researcher, and administrator has one user account, and each audit log entry refers to the user account that performed the action.
- A clinician is assigned zero or more patients (one to many), and each patient is assigned to exactly one clinician.
- Each patient has zero or more assessments.
- An assessment has one recording, at most one transcript, one question response for each question asked, zero or more assessment notes, and zero or more reports.
- A transcript is made up of an ordered list of transcript segments.
- An interference flag belongs to the question response during which it occurred.
- Question responses and transcript segments are linked by time: the segments within a question's period of the recording form the transcript for that question. A question's period starts when the previous question was answered (or at the start of the recording, for the first question) and ends when the patient answers it.

7.3 Annotations

In this document, an **annotation** is any information that is added to an assessment recording to describe it, beyond the recording itself. Annotations are generated automatically or by the clinician, and are then reviewed and corrected by the clinician. They are stored across the entities above:

- **Transcript annotations** (Transcript Segment): the text of each segment, its speaker and language, and its start and end times.
- **Question annotations** (Question Response): the clinician's evaluation of the answer and the note on the question.

- **Behavior annotations** (Interference Flag): moments of possible interference detected by the system or added by the clinician, and whether the clinician kept each one.
- **Session annotations** (Assessment Note): the clinician's notes on the assessment as a whole.

Every annotation also carries metadata describing where it came from, so that clinicians can judge how much to rely on it (NFR14) and so that changes can be traced (NFR8):

- **Source:** whether it was generated automatically or added by a clinician.
- **Verification:** whether a clinician has reviewed it.
- **Original version:** for automatically generated annotations that a clinician has edited, the unedited version.
- **History:** who created or changed it and when, recorded in the audit log.

7.4 Data Dictionary

The table below lists attributes that the requirements in this document depend on. The *Type* column gives the kind of value and, where there is a fixed set, the allowed values. The *Source* column states where the value comes from: the patient-facing site (Patient), the question bank (Question bank), automatic processing (System), the clinician (Clinician), the account holder (User), or the system administrator (Administrator).

Times within a recording must be stored to the nearest tenth of a second or better, so that transcript segments and questions can be synchronized with playback within SYNC_TOL (NFR13).

Attribute	Type	Source	Description
User Account			
login identifier	Text (e.g., email address)	Administrator	Unique name the user signs in with
credential	Protected secret	User	The secret used to confirm the user's identity; never stored or displayed in readable form (NFR15)
role	Choice: clinician, researcher, administrator	Administrator	Determines what information the user may access (NFR5)
Clinician			
name	Text	Clinician	Name of the clinician

Attribute	Type	Source	Description
Patient			
name	Text	Clinician	Name of the child
date of birth	Date	Clinician	Used to interpret results by age and to apply the RETENTION period
guardian	Text	Patient	Name of the patient's guardian
guardian consent	Yes/No	Patient	Whether the guardian has consented to the recording and processing of the patient's assessments
Question			
language	Choice: one of the supported assessment languages	Question bank	Language of the question
assessment type	Choice: matching, repetition, quantifier	Question bank	Type of task
prompt	Media (audio and images)	Question bank	What is presented to the patient
correct answer	Text	Question bank	The correct option; empty for questions that cannot be graded automatically
Assessment			
date	Date and time	Patient	When the assessment was completed
language	Choice: one of the supported assessment languages	Question bank	Language of the assessment
assessment type	Choice: matching, repetition, quantifier	Question bank	Type of task in the assessment
score	Percentage	System	Share of questions evaluated as correct
status	Choice: processing, ready for review, reviewed	System or Clinician	Progress of the assessment: processing while the transcript and interference flags are generated, then ready for review, then reviewed once the clinician marks it complete

Attribute	Type	Source	Description
Question Response			
patient answer	Text	Patient	The option selected by the patient; empty for questions answered aloud
answer audio	Media (audio)	Patient	Recording of the patient's spoken answer, for questions answered aloud (e.g., repetition)
answered at	Recording time	System	The moment in the recording when the patient submitted their answer
evaluation	Choice: correct, incorrect, undetermined	System or Clinician	Whether the answer is correct (FR16)
evaluated by	Choice: system, clinician	System	Who set the evaluation
note	Text	Clinician	Clinician's note on the question (FR17)
Recording			
video and audio	Media (video)	Patient	The recording of the whole assessment
duration	Recording time	System	Length of the recording
Interference Flag			
time	Recording time	System or Clinician	When the event occurred
reason	Text	System or Clinician	What was detected or observed, e.g., the number of faces and the keyword heard
source	Choice: system, clinician	System	Whether the flag was detected automatically or added by a clinician
decision	Choice: pending, kept, dismissed	Clinician	Whether the clinician kept the flag after reviewing it; only kept flags appear in reports
Transcript			
generated by	Text	System	The speech recognition model and version that produced the draft
generated at	Date and time	System	When the draft was produced

Attribute	Type	Source	Description
status	Choice: being generated, draft, verified	System	Progress of the transcript (UC10)
Transcript Segment			
start, end	Recording time	System or Clinician	The period of the recording covered by the segment
text	Text	System or Clinician	What was said
speaker	Choice: System, Child, Parent, Experimenter, Multiple Speakers, Others	Clinician	Who was speaking (FR6)
language	Choice: one or more of the supported assessment languages	System or Clinician	Language or languages spoken in the segment, including segments that switch between languages
original version	Text and times	System	The automatically generated text, speaker, language, and times, kept unchanged after edits
verified	Yes/No	Clinician	Whether a clinician has reviewed the segment (NFR14)
Assessment Note			
text	Text	Clinician	Content of the note (FR18)
Report			
generated at	Date and time	System	When the report was generated, so it can be compared with later changes to the assessment (UC11)
generated by	Reference	System	The clinician who generated the report
file	Document (PDF)	System	The report that the clinician views and downloads
Audit Log Entry			
user	Reference	System	The user account that performed the action
action	Choice: view, change, delete, download, export	System	What the user did

Attribute	Type	Source	Description
item	Reference	System	The patient information that was affected
time	Date and time	System	When the action occurred

Table 3: Data Dictionary

7.5 Privacy and Data Retention

The system stores recordings, transcripts, and health information about children, so the following apply to all data listed above:

- **Access control:** A clinician can only access the patients, assessments, recordings, transcripts, and reports of patients assigned to them (NFR5). Guardians can only access their own child's assessments on the patient-facing site.
- **Encryption:** Recordings, transcripts, and patient records are encrypted in transit and in storage (NFR7).
- **Consent:** A guardian must consent to the recording of assessments and their processing by third-party services before the patient can complete an assessment.
- **Third-party processing:** Audio is sent to external speech recognition and keyword detection services (Whisper and Deepgram). Voice recordings can identify a patient, so these services must be covered by a Business Associate Agreement. Only the audio needed for transcription is sent, without names or other patient details.
- **Exports and reports:** Downloaded reports, transcripts, exported data, and video clips leave the system's protection. Only the assigned clinician may export them, each export is recorded, and exported reports are marked as confidential.
- **Development data:** Real patient data is never to be used in development or testing.
- **Retention:** Patient records, recordings, transcripts, and reports are kept for the RETENTION period, after which they are deleted. Backups follow the same retention period (NFR9).
- **Deletion:** When a guardian withdraws consent or requests deletion, the patient's recordings, transcripts, and assessment data are deleted, including from backups, within DELETE_WINDOW, except where a regulation requires them to be kept.
- **Auditing:** Every view, change, download, and export of patient information is recorded with the user and time, and these records are kept for LOG_RETENTION (NFR8).

- **Compliance:** The system is designed to meet the HIPAA technical safeguards (NFR15). Administrative and physical safeguards, such as staff training and breach notification procedures, are the responsibility of the institutions that use the system and are outside the scope of this project.

8 Non-Functional Requirements

8.1 Usability

NFR1: The system shall be intuitive for users with limited technical fluency.
Rationale: Clinicians' technical fluency varies, and they have little time to learn new software. An intuitive system shortens the learning curve and avoids the need for formal training.
Fit Criterion: In usability testing, at least TASK_SUCCESS_RATE% of clinicians with no prior training on the system must complete each core task (e.g., assign a speaker role, view assessment results, add a patient) without assistance within TASK_TIME minutes.

NFR2: The system shall present all functionality with clear and descriptive labels.
Rationale: Clinicians work under heavy workloads, and they may be unfamiliar with common software icons. Clear labeling lets them recognize what each function does without guessing.
Fit Criterion: In usability testing, at least TASK_SUCCESS_RATE% of clinicians must be able to locate and use each core function on their first attempt, relying only on the labels shown on screen and without assistance.

NFR3: The system shall allow clinicians to return to any previously visited stage of their workflow with minimal effort.
Rationale: Clinicians do not analyze assessments in a straight line. They may go back to review or correct earlier work, so moving between stages should not slow them down.
Fit Criterion: The clinician must be able to reach any previously visited stage of the workflow from the current page in at most MAX_NAV_STEPS navigation steps.

NFR4: The system shall immediately show clinicians the result of each change they submit.
Rationale: Clinicians edit sensitive patient information; therefore, they need quick confirmation that the system received their change and that it was applied correctly. Without immediate feedback, they cannot tell whether their action was registered, which causes uncertainty and repeated attempts.
Fit Criterion: The system must display the updated information to the clinician within SYSTEM_FEEDBACK_TIME seconds after the clinician submits a change.

8.2 Security

NFR5: The system shall only allow a clinician to view the information of patients assigned to them.
Rationale: Patient assessments are sensitive health information. Access must be limited to the clinician responsible for the patient in order to protect patient privacy and meet data protection obligations.
Fit Criterion: The system must only list patients assigned to the logged-in clinician and must not allow a clinician to add or access a patient who is in another clinician's care.

NFR6: The system shall require users to re-enter their credentials after IDLE_TIMEOUT minutes of inactivity.
Rationale: Clinicians may leave their workstation unattended, and their accounts contain highly sensitive information. Automatic sign-out prevents unauthorized people from viewing it.
Fit Criterion: The system must end the user's session and require credentials before displaying any patient information once the user has been inactive for IDLE_TIMEOUT minutes.

NFR7: The system shall encrypt patient recordings and transcripts in transit and in storage.
Rationale: Assessment recordings and transcripts contain private patient information. The annotation service sends video assessments to a third-party transcription service, and the recordings are also kept in the system's storage. Hence, they must be protected in both places so that anyone who intercepts the data or gains access to the storage cannot read it.
Fit Criterion: The system must transmit all recordings and transcripts to third-party services using TRANSIT_ENCRYPTION_STANDARD and must store them using STORAGE_ENCRYPTION_STANDARD.

NFR8: The system shall record who accessed or changed patient information, and when.
Rationale: HIPAA requires a record of activity in systems that contain patient health information, so that unauthorized access can be detected and investigated. Assessment results also inform clinical decisions and reports, so clinicians may need to review how a result changed over time.
Fit Criterion: For every view, change, download, or export of a patient's records, recordings, transcripts, evaluations, notes, annotations, or reports, the system must store the identity of the user, the action performed, the item affected, and the date and time. These records must not be editable by clinicians or researchers and must be kept for LOG_RETENTION.

8.3 Maintainability

NFR9: The system shall regularly back up patient information and assessment recordings.
Rationale: Assessment data is difficult or impossible to recreate, so losing it would waste clinicians' work and could affect patient care.
Fit Criterion: The system must back up all patient information and assessment recordings at least every BACKUP_INTERVAL hours.

8.4 Recording Quality

NFR10: The system shall only accept video assessments in which the speakers can be clearly seen and heard.
Rationale: The annotation tool and behavior detection depend on the recording's quality. Poor audio makes the auto-generated transcript less accurate, whereas poor video quality makes it difficult to detect faces and parent interference. Both instances can lead clinicians to wrong conclusions about a child's assessment.
Fit Criterion: The system must only accept submitted assessments where the recording at least meets the minimum video resolution, MIN_VIDEO_RESOLUTION, and the minimum audio level, MIN_AUDIO_LEVEL.

8.5 Accuracy

NFR11: The system shall raise automatic flags that are accurate enough for clinicians to use them to decide which parts of an assessment need review, with clinicians still making the final judgement.
Rationale: To make things easier, clinicians can use the flags in the assessment overview (FR12) or the interference markers (FR13) to quickly identify which segments need in-depth analysis. Too many missed flags mean certain signals go unnoticed by clinicians, while too many false flags waste clinicians' time.
Fit Criterion: On a set of assessments reviewed by a clinician, the system flags at least FLAG_DETECTION_RATE% of the moments the clinician identifies, and no more than FLAG_FALSE_ALARM_RATE% of its flags are false alarms. Each flag is placed within FLAG_TIME_TOLERANCE seconds of when the moment occurs in the recording.

8.6 Performance

NFR12: The system shall respond quickly enough that clinicians can review assessments without noticeable waiting or lag.
Rationale: Clinicians review many assessments in one sitting, so small delays can add up, especially if they are working on a tight schedule. Jumping to a question also needs to feel instant so clinicians can quickly compare responses.
Fit Criterion: On a connection of at least MIN_BANDWIDTH, the assessment

overview (FR12) loads within OVERVIEW_LOAD_TIME seconds, and selecting a question (FR14) starts video playback at its timestamp within SEEK_TIME seconds. Transcript synchronization during playback is covered by NFR13.

NFR13: The system shall keep the transcript synchronized with the recording during playback and navigation.
Rationale: Clinicians verify transcripts and evaluate answers by matching text to what they see and hear. If the highlighted segment lags behind the recording, or a selected segment starts at the wrong moment, clinicians may attribute speech to the wrong segment or question.
Fit Criterion: For recordings up to MAX_REC_LEN minutes long, the difference between the playback position and the stored start time of the highlighted or selected segment must not exceed SYNC_TOL milliseconds, and playback must begin within SEEK_TIME seconds of the clinician selecting a segment.

8.7 Understandability

NFR14: The system shall make it clear to clinicians which content was generated automatically and which has been verified or edited by a clinician.
Rationale: Clinicians emphasized that automated results are cues for review, not final judgments. To decide what still needs their attention and how much to rely on a result, they need to understand at a glance where each transcript segment, score, and flag came from.
Fit Criterion: In usability testing, at least TASK_SUCCESS_RATE% of clinicians must correctly identify which items on a screen are unverified automated content and which are clinician-verified, without assistance.

8.8 Compliance

NFR15: The system shall meet the technical safeguards of the HIPAA Security Rule for the patient health information it stores and processes.
Rationale: The system stores recordings, transcripts, and assessment results of children, which are protected health information for U.S.-based clinicians. Removing names is not enough to de-identify this data: under HIPAA, full-face images and voice recordings can themselves identify a patient. Clinicians identified HIPAA compliance and control over who can access data as conditions for adopting the system, and institutions must approve tools before clinicians can use them (**?**).
Fit Criterion: Each HIPAA technical safeguard (access control, audit controls, integrity, person or entity authentication, and transmission security) must be mapped to at least one implemented mechanism in the system, and a review of the system against this mapping must find no unaddressed safeguard. Every third-party service that stores or processes patient health information, including recordings and audio, must be covered by a Business Associate Agreement.

8.9 Capacity

NFR16: The system shall store the patients, assessments, recordings, and annotations of a full clinical caseload for every clinician using it.
Rationale: Clinicians may each care for 30 or more patients, and every assessment adds a recording, a transcript, and annotations that must be kept for the full retention period. The system must hold this data without clinicians having to delete older assessments to make room for new ones.
Fit Criterion: With NUM_CLINICIANS clinicians, each with CASELOAD patients who have each completed ASMTS_PER_PT assessments with recordings of up to MAX_REC_LEN minutes, the system must store all of the data and still meet NFR4 and NFR13.

9 AI and Machine Learning Requirements

The system integrates a speech recognition model that was suggested and provided by the capstone team's supervisors. The model was not created by the capstone team.

The model used is OpenAI's Whisper-1, a general-purpose, multilingual automatic speech recognition and translation model. It is accessed through an API and takes audio or text as input (**?**).

9.1 Model Inputs

Individual segments of the assessment's audio file (.wav), each under 25 MB.

9.2 Model Outputs

A JSON object containing the start and end times of each segment, formatted as MM:SS (minutes:seconds), and a text field with the words spoken in the segment.

9.2.1 Example Output

```
[
  {
  "start": "00:00",
  "end": "00:05",
  "text": "Recognized words."
  }
]
```

9.3 Evaluation Metrics

The model is not currently being evaluated. The annotation tool does, however, let speech clinicians edit transcriptions in chunks, viewing the text next to the audio playback, so incorrect auto-generated transcriptions can be corrected.

The model could be evaluated in the future by comparing the original auto-generated transcriptions against the clinician-edited versions and calculating the word error rate.

9.4 Known Limitations

The model's website describes its performance as average, so the following failure modes are expected when it is used in MERLS (**?**).

- **Incorrect transcription or translation:** since the model's overall performance is average, some transcribed or translated text can be wrong. If these errors go unnoticed, clinicians may draw incorrect conclusions about a child's speech abilities.
- **Incorrect transcription of difficult non-English words:** : the model is known to incorrectly transcribe difficult words in languages other than English. This is concerning for MERLS considering the system is used to assess bilingual children.
- **Uneven performance across different languages:** The model generally performs best on languages with a decent amount of training data, such as Spanish, German, Italian, and English. Transcript accuracy may vary by language, and languages with fewer resources may be transcribed less reliably.

Because of these expected failures, MERLS doesn't treat the model's output as final. Clinicians review the auto-generated transcript against the video playback and correct any errors before relying on it.

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR3	–	FR16
SR4	–	FR16
SR6	UC11	FR7
SR3	UC11	FR7
SR6	–	FR17
SR6	–	FR18
SR2	UC10	FR19
SR1	UC8	FR20

SR4	UC8	FR20
SR1	UC8	FR21
SR4	UC8	FR21
SR9	–	FR26
SR10	–	FR26
SR3	–	FR27

Table 4: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR6	NFR2
FR7	NFR4
FR8	NFR3
FR9	NFR5
FR10	NFR4
FR22	NFR1
FR12	NFR11, NFR12
FR13	NFR11
FR14	NFR12
FR16	NFR8
FR16	NFR14
FR17	NFR8
FR18	NFR8
FR19	NFR16
FR19	NFR15
FR20	NFR13
FR21	NFR13
FR26	NFR2
FR27	NFR14

Table 5: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

[Provide a rationale for the decisions made in the documentation. Rationale should be provided for scope decisions, functional requirements, nonfunctioanl requirements, data requirements, and AI requirements. You don't need a rationale for everything, ust the major decisions that are not obvious. You should have traceability to the entities in the document that you are referencing. —TPLT]

12 Likely Changes

LC1: [Give the likely changes, with traceability to the entity that is likely to change. —TPLT]

13 Unlikely Changes

LC2: [Give the unlikely changes. The design can assume that the changes listed will not occur. —TPLT]

14 Appendices

14.1 Appendix A: Additional Materials

This section presents key notes and quotes gathered during the requirements elicitation process for the project.

Several requirements were drawn from research papers describing the system and an experiment conducted by the project supervisor's team in the United States. Although the papers have not yet been released, they describe the entire system and report usability tests with 10 speech clinicians. The following highlights the preferences and feedback participants shared while testing the MERLS system and the SAFA annotation tool.

- Participant 2 expressed frustration with the amount of time it takes to transcribe assessments: "It would be perfect to have this feature, because doing the annotation part is very time consuming."
- Participant 3 disliked transcribing speech assessments and was enthusiastic about saving time: "when you say that it decreases transcription time, I got very excited about having less time in analyzing speech samples."
- Participant 3 also commented on how BeViz displays results: "I would definitely use the chart/graph/percentages." This is in regard BeViz in terms of displaying results.
- Participant 3 further said they preferred having analysis results summarized as scores, summaries, and patterns rather than as a spreadsheet: "quick and helpful, instead of everything in a spreadsheet."
- Participant 5 suggested what the report should contain: "The whole transcripts for all the [assessment] conversations would be useful to include in the report."
- Participant 5 also raised a concern about data privacy: "the challenge will be the HIPAA, and who has access to data, and how the data could be shared with the parents and the educator."
- Participant 7 appreciated the idea of auto-generated transcripts with human verification in the loop: "we can see a real-time transcription when we play the recording, so that we can do the analysis and give feedback."
- Several clinicians emphasized keeping a human in the loop during the assessment analysis portion of the workflow. For example, Participant 9 stated, "[the system should] allow the clinician to make the final judgment; this is a tool that does not replace our judgment."
- Several clinicians found the system straightforward. For example, participant 10 stated: "I think it's straightforward. You can see the video recording, originals, [and] place to write notes."

- Clinicians appreciated that the system covered the different aspects of the speech assessment workflow, particularly the ability to move from overall results to an individual question and review its prompt and response. Participant 10 stated, "you have the 'proof' there to support your clinical decision and your assessment."

- Participants preferred to have results presented in both English and Mandarin side by side, and to have assessments organized meaningfully by semantics, grammar, and phonology. Participant 10 elaborated: "if you have a feature comparing two languages, side by side in the same graph that would probably be helpful to decide differences vs. disorder."

14.2 Appendix B: Development Plan

Development of the system will be split into three phases, with each phase building on the previous one so that the most important goals are completed first. Newer features are then added once the main system is complete and functional. The timing of each phase then follows the milestones outlined in the Development Plan.

1. **Phase 1: Integration.** The first phase focuses on integrating SAFA into the existing MERLS platform, so clinicians can review and annotate assessments without ever having to leave the platform. This includes connecting SAFA's existing transcription and annotation Python modules to MERLS's patient records and recorded assessments, and making sure that patient data stays secure as it moves in between. This phase covers the *Integrate SAFA Into the MERLS platform* and *Protect Patient Data* goals, and will be demonstrated in the proof of concept.

2. **Phase 2: Enhance the SAFA User Interface.** Once SAFA is integrated, the second phase focuses on improving its interface so that speech clinicians can use it with ease, regardless of their knowledge and technical expertise. This involves making the annotation workflow simpler and more intuitive, and requires addressing feedback from clinicians on the current SAFA tool (see Appendix A). This phase covers the *Improve SAFA's Usability* and *Speed Up Speech Annotation* goals.

3. **Phase 3: New Features.** The team will add new features that extend the system beyond SAFA's current capabilities, such as the stretch goals from the Problem Statement. This includes generating patient-facing assessment reports and comparing multiple assessments for the same patient over time. Any features not completed during the course will be documented as future work.

14.3 Appendix C: Values of Auxiliary Constants

Values marked with an asterisk (*) are initial estimates that we will confirm with the supervisor and research team, and will be updated in this document once their values are known. Tested values will be verified during testing of the system.

Constant	Value	Justification
TASK_SUCCESS_RATE*	80%	Slightly above the average completion rate of about 78% found across more than 1,100 tasks in usability tests (**?**); used by NFR1, NFR2, and NFR14.
TASK_TIME*	5 minutes	Long enough to complete a single core task (e.g., add a patient) by a first-time user.
MAX_NAV_STEPS*	3 steps	Any stage of the workflow (patient list, assessment, question, transcript) is within three levels of navigation.
SYSTEM_FEEDBACK_TIME	2 seconds	Keeps the response within the range users perceive as immediate for a saved change.
IDLE_TIMEOUT*	15 minutes	HIPAA requires automatic sign-out after a set period of inactivity but does not specify the period (**?**); 15 minutes is a common choice for applications that display health information.
TRANSIT_ENCRYPTION_STANDARD	TLS 1.2 or higher	The minimum version NIST requires for protecting data sent over a network (**?**). HHS refers to this guideline for securing health information in transit (**?**).
STORAGE_ENCRYPTION_STANDARD	AES-256	The encryption standard approved by NIST (**?**), consistent with HHS guidance for securing stored health information (**?**). Supported by the existing cloud storage (Amazon S3).
BACKUP_INTERVAL	24 hours	Daily backups limit any data loss to a single working day.
MIN_VIDEO_RESOLUTION*	640 × 480 pixels	Minimum resolution at which faces can be reliably detected for interference detection.
MIN_AUDIO_LEVEL*	−30 dBFS average	Minimum speech level at which the transcription model can reliably recognize speech.
FLAG_DETECTION_RATE*	80%	Catches most moments of interference, so clinicians can rely on the flags to decide where to look, while still making the final judgment.

Continued on next page

Constant	Value	Justification
FLAG_FALSE_ALARM_RATE*	20%	Keeps incorrect flags rare enough that clinicians do not stop trusting them.
FLAG_TIME_TOLERANCE*	2 seconds	The existing interference detection pairs a detected face with a keyword heard within 1.5 seconds, so a flag can be placed up to about this far from the moment itself.
MIN_BANDWIDTH*	25 Mbps	A common home or clinic broadband speed, comfortably above what is needed to stream assessment video.
OVERVIEW_LOAD_TIME*	3 seconds	Short enough that clinicians stay focused when moving between assessments.
SYNC_TOL	500 ms	Browsers report the playback position every 15 to 250 ms during playback (**?**), so 500 ms can be met reliably. It is measured against the times stored for each segment, since the accuracy of the speech recognition model's own timestamps is outside the system's control.
SEEK_TIME	2 seconds	Allows for loading the requested part of a recording streamed from cloud storage.
MAX_REC_LEN	30 minutes	Taken from the existing TeleHealth platform, whose requirements set the maximum assessment length to 30 minutes. Integrating SAFA does not change the length of an assessment.
NUM_CLINICIANS*	10 clinicians	A small clinic or research study; can be tested by loading generated records.
CASELOAD*	30 patients	An estimate of one clinician's active caseload, to be confirmed with the research team.
ASMTS_PER_PT*	10 assessments	Several assessment types in each supported language, repeated for progress monitoring.

Continued on next page

Constant	Value	Justification
LINK_EXPIRY	60 minutes	Taken from the existing TeleHealth platform, whose recording links expire after 60 minutes. This keeps a shared link short-lived while lasting through a typical review session; links are renewed while in use.

Table 6: Tested Values of Auxiliary Constants

Policy values describe how long data is kept. They are set in the system's configuration and checked by review, since they cannot be tested within the project's timeline.

Constant	Value	Justification
LOG_RETENTION	6 years	HIPAA requires security documentation to be kept for six years (**?**). Activity records are kept for the same period.
RETENTION*	7 years after the last assessment, or until the patient turns 19, whichever is later	California requires medical records of minors to be kept until at least one year after the patient turns 18, and for no less than seven years (**?**). This rule was also used by the previous TeleHealth project. It is an estimate because the state where clinicians practise has not been confirmed.
DELETE_WINDOW*	30 days	Allows deletion to reach all storage and backups.

Table 7: Policy Values of Auxiliary Constants

14.4 Appendix D: Generative AI Disclosure

14.4.1 AI Tools Used

- Claude Sonnet 5 (Medium)
- ChatGPT-5.6 Luna

14.4.2 How and Where Used

Both models were used in all the sections to do the following:

- Check that each written section answered every question asked and met the rubric requirements

- Improve the wording, clarity and the flow of written sections
- Debug possible LaTeX errors

The models were used to do the following in specific sections:

- Help brainstorm non-functional requirements in section 8, more specifically for security requirements
- Help verify correctness of the FSM diagrams to ensure workflows matched those specified in the research paper in the functional requirements section (Section 6).
- Brainstorm possible non-functional requirements in section 8

14.4.3 Verification of AI-Generated Content

To verify the output of the LLM, each team member went over their section and presented their work to the group, explaining what they had completed since the previous meeting. Each member was able to successfully explain their sections. Additionally, before any pull request was approved, a team member who had not written the section reviewed the work to ensure it was cohesive, accurate, and consistent with the rest of the document.

The factual content of the documents came from the research of the group members. The writer of each section researched, obtained, and documented all referenced sources. After the models suggested wording improvements, the writer re-read the section to confirm that it still conveyed the same point and that the research was not altered.

14.5 Appendix E: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

14.6 Nour

Question 1

Writing the functional requirements was more straightforward than we expected because we had enough resources to understand what our supervisors wanted from the project. They provided us with a couple of unreleased research papers that described the system architecture, clinician workflows, including a study they conducted on 10 clinicians to test the TeLaC (now MERLS) platform. With this information we were able to create a mental model of the MERLS, SAFA, and BeViz modules and what the purpose for each is. Moreover, our supervisors provided us with the GitHub repositories of the annotation tool SAFA and the MERLS system (which is the overall TeleHealth system). That let us write requirements that were realistic in how the system behaves and how it should behave once we integrate SAFA. Additionally, it made it easier to write use cases that reflected the different clinician workflows detailed in the research papers which are, direct analysis, annotation first analysis, and on-demand refinement. These three are the main important workflows and should be enabled by the system very seamlessly. Moreover, after speaking with our supervisors, we also learned which areas they are prioritizing and what the main features of the updated system should be. This helped us determine what requirements could change and what shouldn't change in the system.

Question 2

Creating user personas was my biggest challenge. The part of the system our group is working on is mainly for clinicians, so it was hard to identify different types of users, and we were worried the personas would end up looking nearly identical. Hence, we created different "types" of clinicians who might use the platform and differentiated them by their work environment and capabilities, such as how heavy their workload is and how comfortable they are with technology. This gave us personas that reflect real differences in how the system might be used, which also helped us write non-functional requirements around usability.

Additionally, deciding how detailed each use case should be was also a bit difficult. I wasn't sure where in the workflow each one should start, and simulating the whole workflow from when the clinician enters the system felt like too much. It would have repeated the same steps, such as logging in and selecting a patient in every use case. Hence, I scoped each use case to a single clinician goal and used the preconditions to set the starting point. For example, the clinician is already logged in and has opened a patient's assessment. I also kept the steps focused on what the clinician does and how the system responds, and left out screen-level details, since the interface may change during the redesign.

Question 4

I took SFWRENG 4AL3, the applications of machine learning course. It helped me understand the kinds of models our supervisors may have used when building the SAFA tool, and I am comfortable with the machine learning terminology in our documents, such as word error rate and model inputs and outputs. Although we did not develop the model and it is not currently being evaluated, I understand how the audio files are segmented and sent to the model for transcription, and I can reason about its known limitations, such as weaker performance on non-English words. This is relevant because our users work with bilingual children. The course also taught me why disclosing AI use matters. We learned about the negative effects of large language models, including their large carbon footprint, and about the importance of being transparent about how they were used in our documents.

Question 5

Team management is extremely important for completing the project successfully. Our work is split across documents for different milestones, and each member's section has to fit with the others, so we need a clear picture of who is doing what and when it is due. Having good team management lets us stay on top of tasks, stay on the same page, and catch problems early on. Moreover, it means regular meetings where each member explains what they have completed since the last one, and discussion regarding the next project steps. This is important considering we are balancing the project along with our other courses. Furthermore, considering we are building on an existing system with already busy supervisors, we need to communicate regularly with them and record their feedback during meetings. Without this, it would be easy for sections to become inconsistent with each other

and for team members to be lost or have a different understanding of the project vision and next steps.

Additionally, understanding of how SAFA uses AI to create transcriptions is also important. Integrating SAFA into the existing MERLS system is the main focus of our project, so we need to understand how the audio is sent to the model, what the model returns, and where its transcriptions may be wrong, so we can design the system around this.

Question 6

For team management, one approach is keeping a planner or schedule that lists upcoming deadlines, notes, and future meetings. It keeps the team on track and helps us balance the project with other courses. Another is writing meeting minutes after each meeting. Minutes help the team manager see everyone's tasks and what still needs to be done, so that work is finished on time and with good quality. I chose to pursue team management because I have experience with group projects and prefer to know what is happening at all times during a project.

As for understanding how SAFA uses AI, one approach is reading the documentation for the speech recognition model and for SAFA, as well as going through SAFA's code. Another is asking our supervisors to walk the team through how transcriptions are generated and where they tend to fail. I will pursue this skill as well, mainly through the documentation and code review, because my machine learning course gave me a background in AI systems and their typical pitfalls, which should help me read the code and judge where the transcriptions may be unreliable.

14.7 Javayria

Question 1

Although this deliverable was definitely more difficult and time-consuming than the previous one, the research papers we had access to were very helpful in showing us what the system needed to do from the very start. We were also given very clear "must-haves" from both our supervisor (Dr. Irene Yuan) and the researchers we are working with. They stressed from the beginning that their main goal was integrating the two systems (TeleHealth and SAFA) and redesigning the interface, and that anything beyond that would be a nice-to-have. This made it much easier to decide what belonged in our core requirements and what should be marked as a stretch goal. For example, clinicians in the study suggested features such as visualizing results by clinical domain and automatically detecting grammatical errors. Because we knew our priorities, we could include these as stretch requirements without letting them expand our scope.

The research papers also already contained feedback from ten clinicians who had tested the platform, which was a great way for us to gather requirements without having to interview clinicians ourselves within the past few weeks (which were very busy due to our courseloads). Their direct quotes gave us a clear, traceable source for our stakeholder requirements. The

papers also described the three clinician workflows (direct analysis, annotation-first analysis, and on-demand refinement), which gave us a natural structure for our use cases and state diagrams. Having the previous TeleHealth team's SRS and code as a reference helped us see what had already been built, so we could focus our requirements on what the integrated system still needs.

Question 2

One of the main pain points was getting the information we needed to write our requirements. Our project is unique in that it integrates two existing tools, so we needed access to both repositories to understand what was already built. However, delays in responses from our supervisors meant we did not receive access until much later into writing the deliverable. In the meantime, we had to start writing requirements based on the research papers alone, which made it difficult to separate the requirements of the overall system from the ones our own work needed to meet.

Since our project builds on the work of others, we also spent a significant amount of time learning what the existing systems already do and how our work needs to fit into them. For example, the previous TeleHealth team built the interface that records children's assessments. We were not sure whether to include requirements for handling that data, since it was not our work, but it would still be part of the system we would deliver at the very end. We resolved this by deciding that any requirement the final system must meet belongs in our SRS, even if another team previously contributed to that part. This turned out to be very important, because we found out that the existing platform does not enforce authentication or data privacy, and our supervisor stressed that the integrated system must protect patient data and follow HIPAA. Once we had access to the repositories, we compared our requirements against the existing code and the previous team's SRS, and updated the ones that did not match how the system actually works.

Question 3

TO-DO

Question 4

Security and data privacy are a major part of this capstone, and I am currently taking COMPSCI 4CR3: Introduction to Cryptography. Our system handles assessment recordings, transcripts, and reports. These are sensitive health information, as patients' faces and voices count as identifiers under HIPAA. Any sensitive information sent over a network has to be encrypted so that attackers cannot read or tamper with it in transit. It also needs protecting while it is in storage. We learned about modern ciphers such as AES, along with key management and secure communication protocols. This can help our team choose suitable standards for encrypting patient data and understand why they are secure, not just how to use them. However, we would first need to work out where in the system data needs to be

encrypted, for example when it is uploaded, stored, or shared through download links. We would also need to plan how to decrypt it efficiently when clinicians retrieve it, so that video playback stays responsive. Not all data needs this level of protection, so we will also have to evaluate which parts of the system are worth guarding this way.

Question 5

Our capstone involves taking a proof of concept and turning it into a software that clinicians can rely on, so the team will need to learn a lot beyond what the existing systems use in order to implement the backend. Currently, TeleHealth is a Node/React web application that stores its data as JSON files, while SAFA is a Python desktop application whose transcription model runs as a separate script, and its output has to be uploaded into SAFA by hand. To build one integrated system, we will need to learn how to connect services written in different languages and run long tasks like transcribing a 30-minute recording in the background. We will also need to track each assessment's progress through the pipeline so clinicians stay in control of every stage. One member will focus on this service integration and pipeline design. Another member will learn how to deploy and evaluate Whisper efficiently, since children's speech and recordings in more than one language are known weak points for speech recognition.

Since the current system has no database, I will focus on learning database design so we can properly link patients, assessments, transcript segments, annotations, and reports, and keep the original AI transcript alongside clinicians' edits. Because the existing platform also has no login sessions or access control, one other team member will focus on authentication, encryption, audit logging, and the HIPAA technical safeguards. As a team, we will also need to learn browser media handling to rebuild SAFA's synchronized video and transcript playback in React, as well as the PDF report generation. Finally, we all need enough domain knowledge of speech-language pathology to understand how clinicians conduct and interpret assessments, so that the workflows we design actually make sense for their daily work.

Question 6

TO-DO

14.8 Catherine

Question 1

I feel like because a lot of work for this deliverable had already been laid out in our Problem Statement and Development Plan, as well as from the many meetings we had with our supervisors and TA, I didn't feel like I was starting from scratch anymore. Having already written about the proof of concept workflow and our main risks, I had a pretty good idea of who the users were and how they would move through the system, as well as how our system was gonna work and what exactly we needed to make it work. Creating the system context diagram also helped more than I expected, since it forced me to pin down exactly who

interacts with the platform and what external systems we'd be depending on, before writing any requirements. Another thing that went well was how directly some of the stakeholder feedback translated into requirements. For example, clinicians wanted results organized by specific domains such as grammar, and they also had concerns about children with limited attention. Both of these ended up becoming our functional requirements.

Question 2

One of the biggest pain points for me was understanding and visualizing how use cases, functional requirements, and non-functional requirements all related to each other. At first I wasn't sure if they were supposed to overlap or reference each other directly, and it was easy to blur the lines, e.g. writing a functional requirement that read more like a use case, or a non-functional requirement that just restated a functional one. What helped was going back to the template and realizing that use cases describe what the user is trying to do, functional requirements break that down into single and verifiable "shall" statements, and non-functional requirements describe how well the system does it, as stated many times throughout our lectures. The traceability tables finally connect them all together, which also helped me understand the concept a lot better.

Question 4

One course that I remember taking was 2AA4, Intro to Software Development. It was a very technical course, but it also involved a lot of requirements and planning when trying to develop a solution slash program to solve a problem. In the deliverables we wrote in the class, I clearly remember having to iterate multiple times over our idea as we failed to consider certain edge cases or requirements to the problem we were facing, which was an island, explore-a-maze type game. Looking back, with more planning, a lot of those iterations could have been avoided, which is exactly what this deliverable is trying to do for our capstone. Thinking through edge cases in our use cases and discussing alternative flows can help us define early on what our system needs in terms of requirements. This will be especially important once we start integrating SAFA, which was built as a standalone tool, into the existing MERLS platform, since there are a lot of moving parts and missing a requirement now could mean having to go back and rework large parts of the system later.

Question 5

Two areas that our team will need to learn more about will firstly be front-end development and making sure that the user experience is as smooth and intuitive as possible, and secondly more on security and how we can set up our system to be more secure. On the front end, a big part of our project is bringing SAFA's interface into the MERLS web platform and making it easier for clinicians to use, so the team will need to be comfortable building interfaces with frameworks like React and potentially using frontend component libraries. Security seems like a pretty large problem for our project as some of the largest issues such as data and

privacy leaks could occur if not handled correctly, especially since our system handles video recordings and records of children. This likely means getting more comfortable with cloud storage services such as AWS, including how to encrypt data and control who has access to it so that clinicians can only see their own patients. These are the two areas I will be focusing on for the team.

Question 6

For front-end development, one approach is to follow how different fullstack projects and tutorials work online, or using React's official documentation or AI as a direct tutor. Another is to learn by working directly in the existing SAFA and MERLS code, as there's some frontend work already done. I believe as a team, we'll likely do a mix of everything, since there's not really a one-size-fits-all approach. I personally learn best by doing, so I'd likely focus on working directly in the existing code, and refer to tutorials or documentation whenever I get stuck.

For security, one approach is to go through AWS's documentation and free training on topics like encryption and access permissions, and another is to build a small prototype on AWS's free tier that stores sample recordings securely and test who can access them. I plan to do the prototype, since actually setting it up would give me a solid understanding of the security measures needed. It would also show me where things can go wrong much better than just reading about it.